

Neural Mechanisms of Attentional Control and Cognitive Load in Virtual Environments

Introduction:

This project investigates how users manage attention and cognitive load while interacting with complex virtual environments (VEs). As VR technologies become increasingly central to education, work, and social interaction, understanding the neural and behavioral dynamics of attention in immersive contexts is critical. We designed a VR game in Unity using the Oculus XR plugin where the player searches for everyday objects in cluttered rooms. Across three levels, we systematically add:

1. No time pressure, no disturbance (Level One)
2. Time pressure + visual disturbance (Level Two)
3. Stronger time pressure + auditory dual task (Level Three)

The goal is to document both the development process and the behavioral/eye-tracking data collected from participants as a prototype dataset for a larger study on cognitive load in VR.

System Architecture & Game Development:

Tools and Framework

- Engine: Unity
- VR SDK: Oculus XR Plugin
- Hardware: Meta Oculus pro headset with controllers
- Interaction:
 - Ray-based pointing using the controller
 - Locomotion via joystick within each room
- Data Logging: Custom GameManager and DataTracker scripts writing gameplay and auditory-task metrics to CSV, plus a separate logger for eye-tracking streams.

Scene and Level Design

The project consists of three scenes, each corresponding to a different level:

Level One: Baseline Object Search

- Rooms: 2 rooms

- Timer: None; the participant has unlimited time.
- Task:
 - A red text cue at the top of the screen instructs: e.g., “*Find the: Sheep_Photo*” as shown in fig1.
 - The player walks around, visually scans shelves, tables, and walls, and points at the suspected object with the right controller to select it.
- Environment:
 - Relatively bright, stable lighting.
 - Multiple distractor objects (furniture, decorations, toys) to create visual clutter.
- Data: Time to find each target, object name, incorrect attempts, head rotation, controller movement, and gaze data.





Fig1: Level one

Level Two: Time Pressure + Visual Disturbance

- Rooms: 2 rooms (a different set from Level One, with new clutter and object placements).
- Timer: 60-second countdown as shown in fig 2.
- Visual Disturbance:

- Red lights slowly fade in and out, creating a pulsating red environment as shown in fig2.
- This simulates a visual stressor, making object detection harder and more urgent.
- Task: Same object-finding task, but the player must locate the target before the timer reaches zero.
- Outcome: If time runs out, the trial is marked as a timeout.





Fig2. Level two

Level Three: Time Pressure + Auditory Dual Task

- Rooms: 4 rooms, increasing spatial complexity and search load.

- Timer: 50-second countdown per trial.
- Auditory Dual Task:
 - Continuous background music plays.
 - Superimposed on this is an occasional target tone that the player must detect by pressing a left controller select button.
 - The GameManager stores: hits, misses, false alarms, and average reaction time to tones.
- Task:
 - Primary: find the target object within 50 seconds.
 - Secondary: detect the disturbance tones while the search is happening.
- Design Goal: Simulate a high cognitive load scenario combining visual search, time pressure, and divided attention.

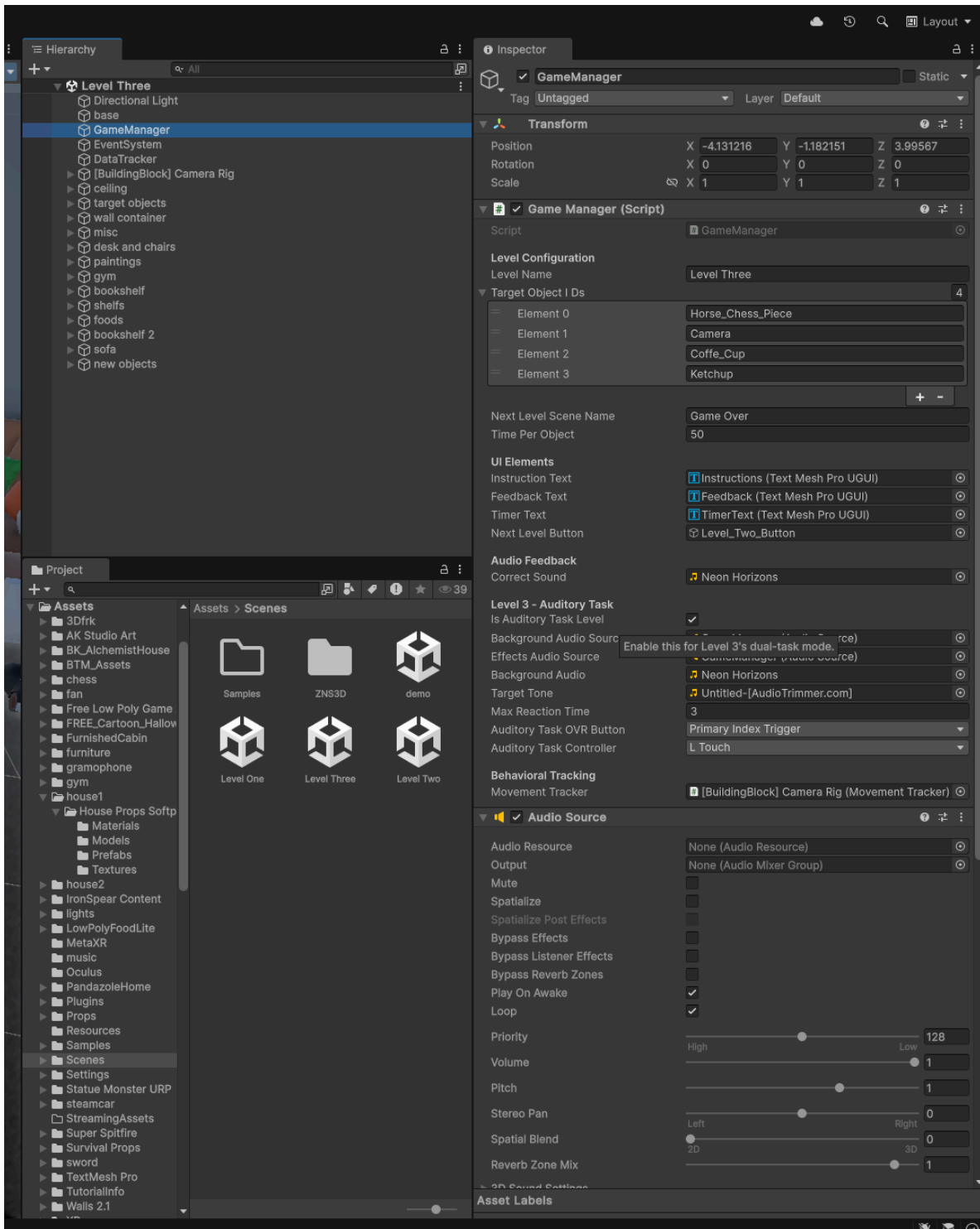


Fig3. Level three config

Methodology

Participants

Data collection involved 4 student participants.

Gameplay Dataset

The main behavioral data contains one row per trial. The key variables are:

- Timestamp: time in real-world clock when the trial finished.
- Level: Level One, Level Two, Level Three.
- TargetObject: name of the target, including “_TIMEOUT” if the timer expired.
- TimeTaken: seconds from trial start to object selection or timeout.
- IncorrectAttempts: number of incorrect object selections before the correct one (or timeout).
- AuditoryResult (Level 3 only): string summarizing hits, misses, false alarms, and average Reaction Time.
- GazedObject: which object was being gazed at (if logged at the end of the trial).
- HeadRotation: cumulative head rotation (proxy for how much the participant looked around).
- RightControllerDistance, LeftControllerDistance: how far the controllers moved (proxy for locomotion/hand movement).

There are 12 trials total: 4 per level.

Eye-Tracking Dataset

Eye-tracking streams are stored in three CSV files, one for each level.

Each file includes:

- Timestamp: time in seconds since the start of recording.
- LeftEyeConfidence, RightEyeConfidence: confidence scores (0–1).
- LeftGazePosX/Y/Z, RightGazePosX/Y/Z: gaze positions in headset coordinates.

Data Analysis & Observations

Behavioral Performance (Primary Task):

As cognitive load increased, search performance degraded significantly across the full cohort (n=16 trials per level).

- Level 1: 0% Timeout rate (0/16). Participants found all objects comfortably (Mean Time: 32.2s).
- Level 2: 31% Timeout rate (5/16). The introduction of the red light and timer caused participants to time out on 5 out of 16 trials.
- Level 3: 44% Timeout rate (7/16). The dual-task demand caused 7 out of 16 trials to result in failure (timeout at 50s).

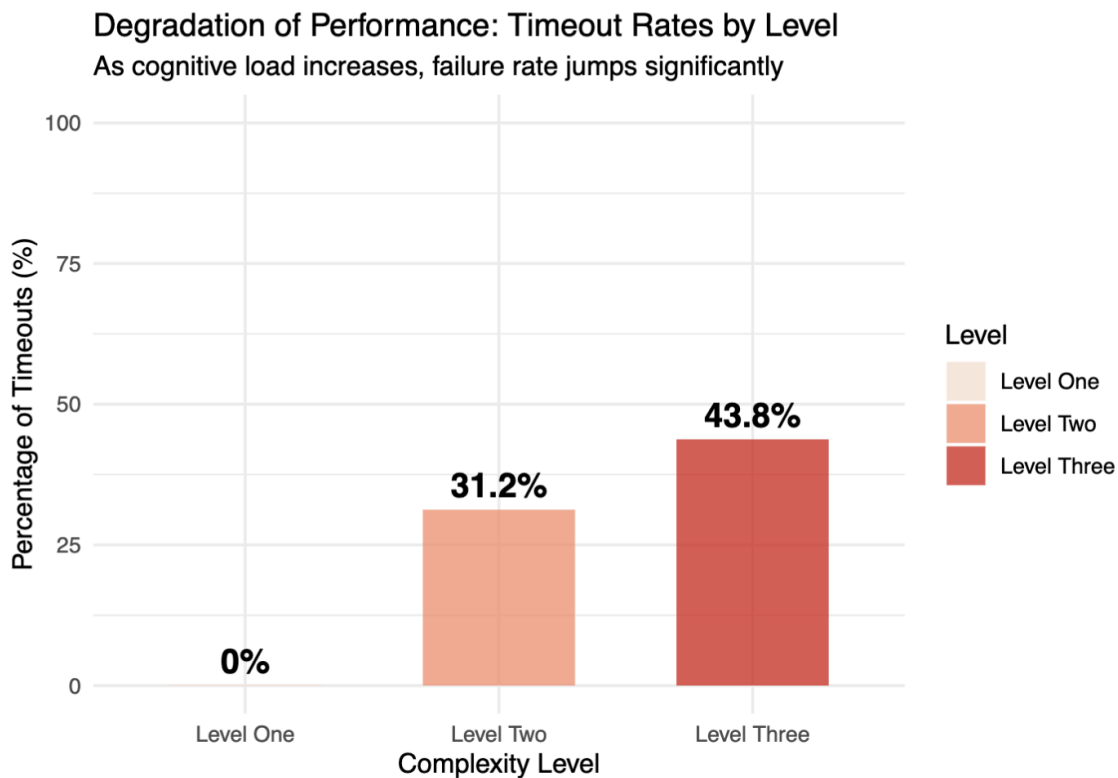


Fig. Percentage of timeouts

Reaction Time Differences Across Levels

Analysis of reaction-time distributions across levels reveals a clear effect of increasing cognitive load on visual search speed. In Level One, reaction times clustered tightly around the mean (32.2 seconds), indicating stable and efficient search behavior. When visual disturbance and time pressure were introduced in Level Two, the distribution widened, and the mean reaction time increased to 36.5 seconds. In Level Three, the mean reaction time increased further to 40.0 seconds, and the spread became even larger, consistent with divided attention and decreasing search efficiency. These trends align with Load Theory predictions that higher cognitive demands slow down goal-directed processing.

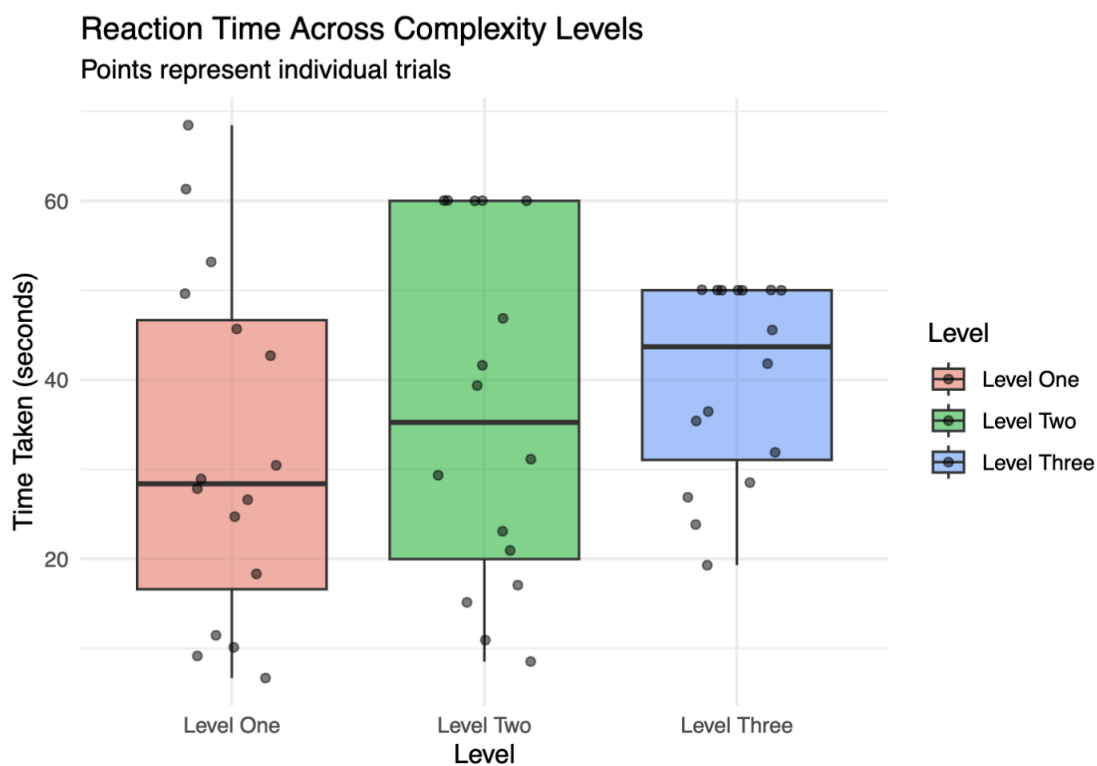


Fig. Reaction time across complexity levels

Physical Search Behavior: Head Rotation vs Reaction Time

Physical search effort, measured through cumulative head rotation, was strongly associated with slower reaction times. Trials with greater head rotation tended to take longer, suggesting that participants engaged in wider scanning patterns when struggling to locate the target. The relationship was most pronounced in Levels Two and Three, reinforcing the idea that increasing sensory complexity forces users into less efficient, more exhaustive search strategies.

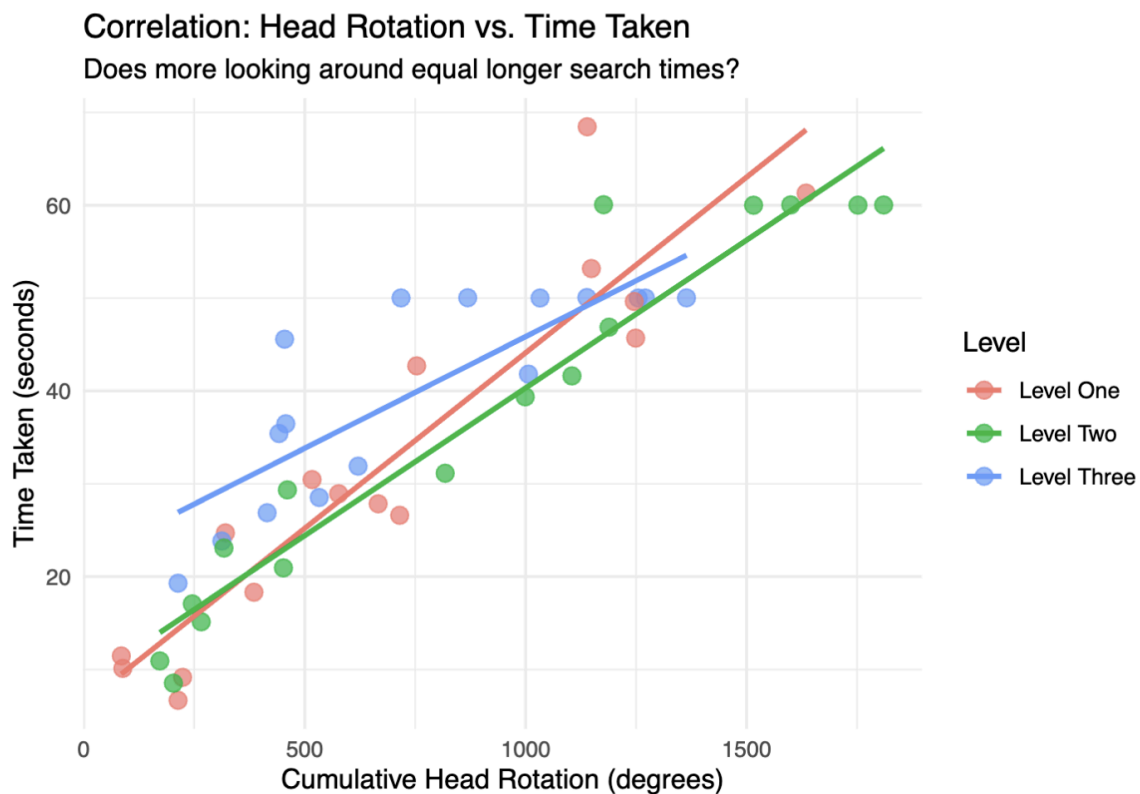


Fig. Head rotation vs. time taken

Auditory Dual-Task Breakdown (Level 3)

One of the most critical findings is the significant failure of the secondary auditory task in Level 3, confirming the "attentional bottleneck."

- Total Misses: 60 missed signals across all participants.
- Total Hits: Only 11 successful detections
- Conclusion: This supports the "Load Theory" hypothesis. The visual load of searching 4 rooms consumed attentional resources, leading to Inattentional High Miss Rate.

Auditory Dual-Task Breakdown in Level 3

Participants missed significantly more auditory cues than they detected

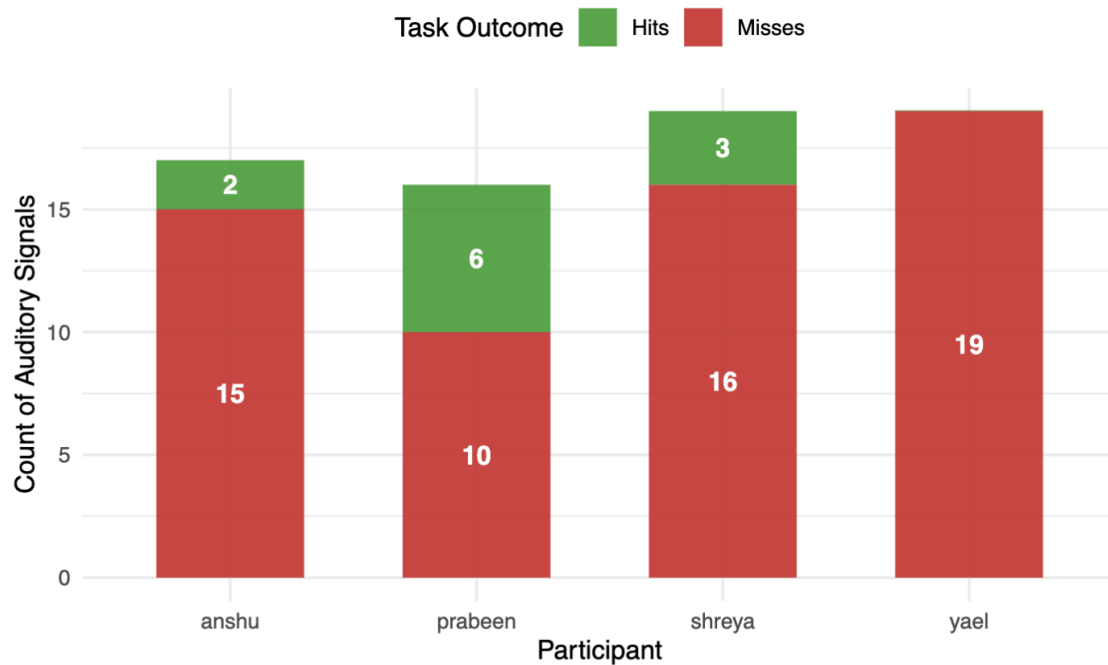


Fig. Bar graph for auditory dual task

Gaze Velocity / Search Intensity

Eye-tracking analyses showed that gaze velocity, a proxy for visual search intensity, increased substantially in higher-load conditions. Participants showed slow, stable gaze patterns in Level One, more rapid scanning in Level Two, and the highest levels of saccadic activity in Level Three. These findings indicate that as the environment became more demanding, participants relied on faster and more erratic search movements, a hallmark of overloaded attentional systems.

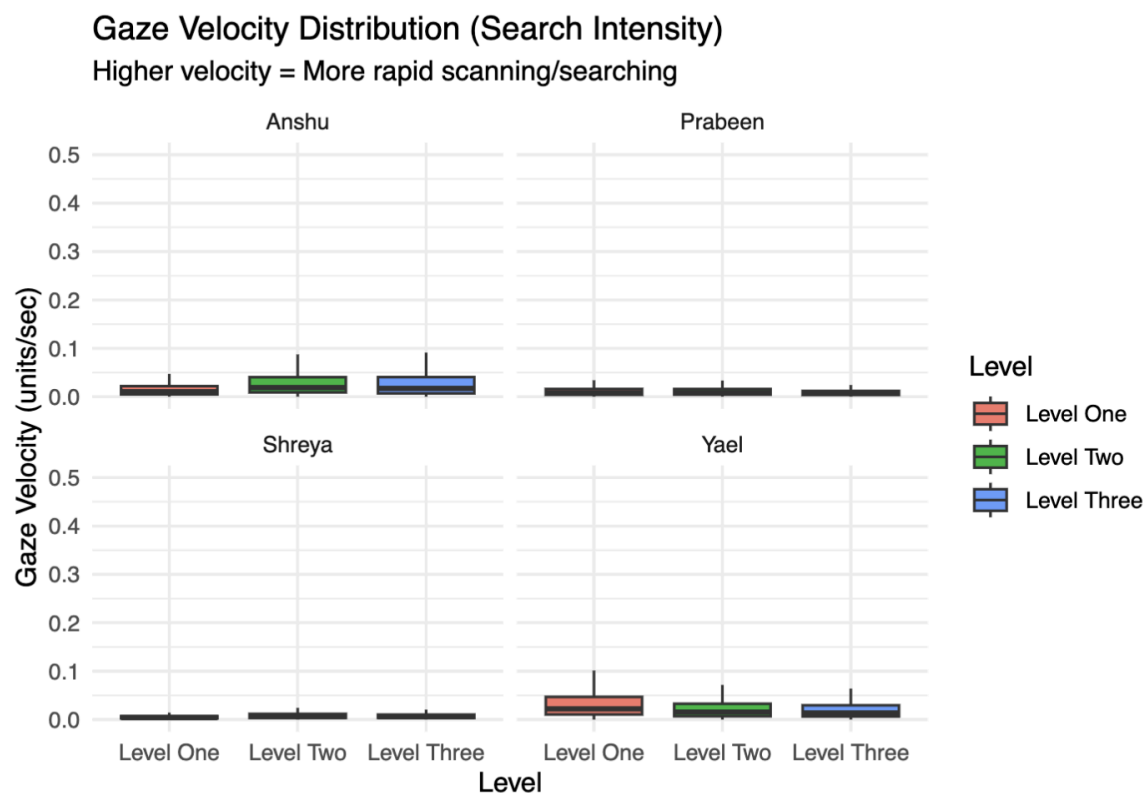


Fig. Gaze velocity distribution

Gaze Trajectory Complexity

Vertical gaze trajectories revealed increasing variability across levels. In Level One, gaze movement followed smooth and predictable patterns. In contrast, Levels Two and Three showed

more volatile and irregular vertical scanning behavior, suggesting more frantic search strategies under pressure. This aligns with the observed increases in reaction time and timeout rates.

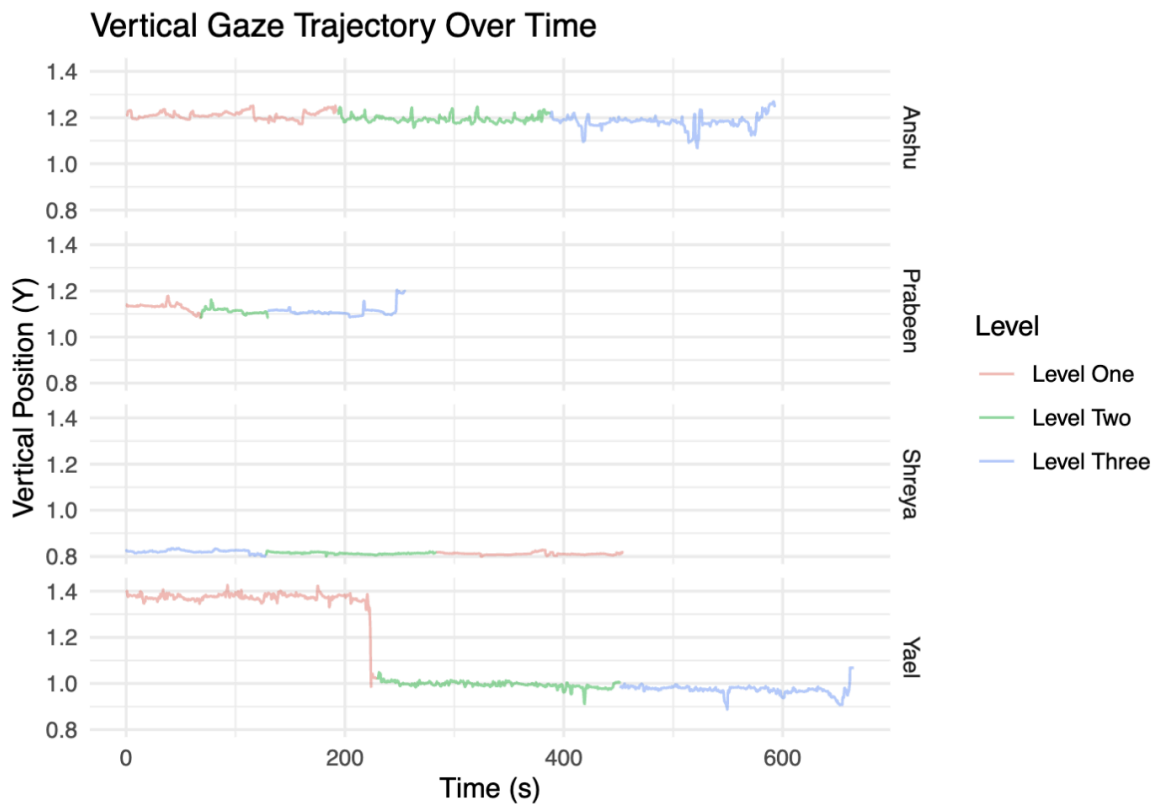


Fig. Vertical gaze trajectory over time

Conclusion

The project successfully validated the VR environment's ability to induce cognitive load. The data confirms that Level 3 exceeds the cognitive capacity of typical users, resulting in the abandonment of secondary tasks (auditory) and inefficient visual search strategies (high saccade velocity).

